

9.4: 12) $\sum_{k=0}^{\infty} [1/2(.2)^k + 3/2(.8)^k]$

$= 1/2 \sum_{k=0}^{\infty} (.2)^k + 3/2 \sum_{k=0}^{\infty} (.8)^k \leftarrow \text{These are geometric series}$
 $a=.1, r=.2 \quad a=1, r=.8$

$= 1/2 \left(\frac{1}{1-.2} \right) + 3/2 \left(\frac{1}{1-.8} \right) = 1/2 \left(\frac{10}{8} \right) + 3/2 \left(\frac{10}{2} \right) = \underline{\underline{65/8 = 8.125}}$

14) $\sum_{k=0}^{\infty} \frac{2-3^k}{6^k} = \sum_{k=0}^{\infty} \frac{2}{6^k} - \sum_{k=0}^{\infty} \frac{3^k}{6^k}$

$= 2 \sum_{k=0}^{\infty} \left(\frac{1}{6} \right)^k - \sum_{k=0}^{\infty} \left(\frac{3}{6} \right)^k \leftarrow \text{geometric series}$
 $a=1, r=1/6 \quad a=1, r=1/2$

$= 2 \left(\frac{1}{5/6} \right) - \frac{1}{1/2} = \underline{\underline{2/5}}$

• 20) $\lim_{n \rightarrow \infty} \frac{n^3}{n^3+1} = 1$ so the series diverges by the Divergence Test

24) $f(x) = \frac{x^2}{\sqrt{x^2+4}}$ is continuous & positive for $x \geq 1$,
 but $f'(x) = \frac{4}{(x^2+4)^{3/2}} > 0$ so we can't use the
 integral test.

$\lim_{n \rightarrow \infty} \frac{x^2}{\sqrt{x^2+4}} = 1$ so the series diverges by the
 divergence test.

$(f'(x) = -\frac{1}{3(x+10)^{4/3}})$

• 36) $f(x) = (x+10)^{-1/3}$ is cont'd & positive & $f'(x) < 0$ for $x \geq 1$.

$\int f(x) dx = \int u^{-1/3} du = \frac{u^{2/3}}{2/3} + C = \frac{3}{2}(x+10)^{2/3}$

Let $u = (x+10) \Rightarrow \frac{du}{dx} = 1 \Rightarrow du = dx$

$\lim_{b \rightarrow \infty} \left[\frac{3}{2}(x+10)^{2/3} \right]_1^b = \infty - \frac{3}{2}(1+10)^{2/3} = \infty$

9.4 29) $f(x) = \frac{x}{e^x}$ is cont^g, positive, & $f'(x) = \frac{(1-x)}{e^x} < 0$
for $x \geq 1$. $\int_1^{\infty} f(x) dx = -x\bar{e}^x + \int \bar{e}^x dx = -x\bar{e}^x - \bar{e}^x = -\bar{e}^x(x+1)$

$$\left[\begin{array}{l} \int u dv = uv - \int v du \\ u = x \quad \frac{dv}{dx} = \bar{e}^x \Rightarrow v = \int \bar{e}^x dx = -\bar{e}^x \\ \frac{du}{dx} = 1 \Rightarrow du = dx \end{array} \right]$$

$$\lim_{b \rightarrow \infty} [-\bar{e}^x(x+1)]_1^b = \lim_{b \rightarrow \infty} -\bar{e}^x(x+1) + \bar{e}^1(2) = \frac{2}{e}$$

So the series converges.

9.5 9) $\lim_{n \rightarrow \infty} \frac{\frac{1}{(k+1)!}}{\frac{1}{k!}} = \frac{k!}{(k+1)!} = \frac{1 \cdot 2 \cdots k}{1 \cdot 2 \cdots k \cdot (k+1)} = \frac{1}{k+1}$ & $\lim_{x \rightarrow \infty} \frac{1}{x+1} = 0 < 1$

So the series converges by the Ratio Test.

10) $\frac{2^{\frac{k+1}{2}}}{k!} = \frac{2^{\frac{k+1}{2}} k!}{2^{\frac{k}{2}} (k+1)!} = \frac{2^{\frac{1}{2}}}{k+1}$, $\lim_{x \rightarrow \infty} \frac{2}{x+1} = 0 \Rightarrow$ the series converges

13) $\frac{2^{k+1}}{(k+1)^{k+1}} \left(\frac{k^k}{2^k} \right) = \frac{2^{k+1}}{k+1} \frac{k^k}{(k+1)^k} = \frac{2}{k+1} \left(\frac{k}{k+1} \right)^k \Rightarrow \lim_{n \rightarrow \infty} \left(\frac{n}{n+1} \right)^n = \bar{e}^{-1}$

$$\left[\lim_{n \rightarrow \infty} \left(\frac{n+x}{n} \right)^n = e^x \right]$$

So then,

$$\lim_{k \rightarrow \infty} \left(\frac{2}{k+1} \right) \left(\frac{k}{k+1} \right)^k = 0 \cdot \bar{e}^{-1} = 0, \text{ so the series converges.}$$

14) $\lim_{k \rightarrow \infty} \left(\frac{k}{k+1} \right)^k = \bar{e}^{-1}$, so then $\frac{(k+1)!}{(k+1)^{k+1}} \cdot \frac{k^k}{k!} = \left(\frac{k}{k+1} \right)^k \xrightarrow{k \rightarrow \infty} \bar{e}^{-1}$ so the series converges

19) $\sqrt[k]{\frac{4k^3+k}{9k^3+k+1}} \Rightarrow \lim_{k \rightarrow \infty} \frac{4k^3+k}{9k^3+k+1} = \frac{4}{9} < 1 \Rightarrow$ the series converges

20) $\left\{ \frac{k+1}{2k} \right\}_{k=1}^{\infty} \rightarrow \frac{1}{2} < 1$ so the series converges by the root test

22) $\left\{ \left(1 + \frac{3}{k}\right)^k \right\}_{k=1}^{\infty} \rightarrow e^3 > 1$, so the series diverges "

23) $\left\{ \left(\frac{k}{k+1}\right)^{2k} \right\}_{k=1}^{\infty} \rightarrow e^{-2} < 1$ so the series converges "

24) $\lim_{k \rightarrow \infty} \frac{1}{\ln(k+1)} = 0 < 1$ so the series converges "

25) $1 + \left(\frac{1}{2}\right)^2 + \left(\frac{1}{3}\right)^3 + \dots = \sum_{k=0}^{\infty} \left(\frac{1}{k}\right)^k$. $\lim_{k \rightarrow \infty} \frac{1}{k} = 0 < 1$
so the series converges "

27) $a_k = \frac{1}{k^2+4} < b_k = \frac{1}{k^2}$ and $\lim_{k \rightarrow \infty} b_k = 0$
so that $\lim_{k \rightarrow \infty} a_k = 0$ by the comparison test

28) $\frac{\frac{k^2+k-1}{k^4+4k^2-3}}{\frac{1}{k^2}} = \frac{k^4+k^2-k^2}{k^4+4k^2-3} \quad \lim_{k \rightarrow \infty} \left(\frac{k^4+k^2-k^2}{k^4+4k^2-3} \right) = 1$

so the series converges by the comparison limit test since $\{1/k^2\}$ converges